Codebook

Aesthetics, Narrative, and Gameplay in AAA vs. Indie Games: What Gamers Talk About in Steam Reviews?

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Background Information

This study examines how players evaluate video games by analyzing user-generated reviews on Steam, with a focus on differences between **AAA** (high-budget, large-studio) and **Indie** (independent, lower-budget) games. Given their distinct production conditions and creative approaches, players may mention different aspects of the gaming experience more when reviewing these two production contexts.

To capture these differences systematically, this codebook organizes the content of these reviews into three key dimensions, including **aesthetics, narrative,** and **gameplay**. Each dimension consists of specific **sub-elements**. The coding scheme is designed to identify which dimensions and sub-elements are mentioned most frequently by players, allowing for a comparative analysis of how these dimensions differ in their frequency of mention between AAA and Indie game reviews.

Research Questions and Hypotheses

- **RQ1:** Which dimension (aesthetics, narrative, or gameplay) is most frequently mentioned by players in Steam reviews of AAA and Indie games?
    - **H1:** Aesthetic dimension is mentioned more frequently in reviews of AAA games than in reviews of Indie games.
    - **H2:** Narrative dimension is mentioned less frequently in reviews of AAA games than in reviews of Indie games.
    - **H3:** Gameplay dimension is mentioned more frequently in reviews of AAA games than in reviews of Indie games.
- **RQ2:** Within the following three dimensions (aesthetics, narrative, and gameplay), which specific sub-elements are most frequently mentioned by players in AAA and Indie game reviews?
- **RQ3:** How do reviews of different game genres differ in how frequently the three main dimensions are mentioned?
- **RQ4:** What is the relationship between played hours and the mentioning of the three main dimensions?

General Information for Coders

Coders will analyze a list of 900 randomly selected reviews in total for 10 games (five AAA, five Indie), drawn from the “Top 100 Most Played Games” list as of April 14 (by current players), 2026 (Steam, n.d.). 100 reviews (around 11% of the sample) will be

repeatedly coded by all four coders to test for inter-coder reliability. Both the production contexts and genres have been identified in advance. The coders will be provided with a list of game reviews across five different genres, including five AAA and five Indie games for each genre. Coders are told the genre of the game and whether it is Indie or AAA. The latest reviews from January 1, 2026 to April 1, 2026 are coded.

Before the actual coding, all coders will have a training session together to discuss and arrive at a common understanding of the codebook as well as the selected ten games. Afterwards, the coders will read through the codebook together to discuss and clarify any uncertainties in the codebook. The coders will then read the game descriptions on Steam and watch clips of YouTube videos displaying actual gameplay sessions for all ten games together.

For the actual coding, coders will be provided with an Excel sheet containing all the necessary information they need to code, including Steam game description, administrative elements (Steam code of the game, review ID, hours played, post date), and full texts of the extracted Steam reviews. Please read the game description thoroughly first to establish a foundational understanding of the game. The reviews should be coded individually in the random order provided in the Excel.

The definitions for the three main dimensions in the codebook (aesthetics, narrative, gameplay) are extracted from the book (Egenfeldt-Nielsen et al., 2015). However, since there is limited literature studying Steam reviews with content analysis, the coding questions for both the main dimensions and the sub-elements are derived by the researchers based on literature and empirical observations of Steam reviews in pilot test (Lin et al., 2018).

Step-by-Step Coding Rules

- Before the coding process:
    - A training session is organised for all coders.
        - In this training session, you will go through this codebook step-by-step to ensure that you and other coders have gone through the codebook in detail.
        - You will read the game description and watch short gameplay videos together from YouTube of all the games that you will analyze the reviews for.
- During the Coding Process:

- Read the Steam game description to gain a basic understanding of the context of the game. These are provided to coders together with the reviews.
- Be discreet when coding for administrative variables, including review ID, game genre, AAA or Indie, post date, and hours played. Avoid typos especially in coding for the review ID.
- Keep the variable definitions and research questions in mind.
  - Think about the keywords listed for each variable, but also consider the context of the words and phrases within the review based on the description provided.
  - The key words are to guide you, not to provide clear yes or no answer options of what must be included.
  - Also, consider similar terms to those stated under Keywords and do not just look for exact matches.
- For any words, slangs, or abbreviations that are not clear to the coder, search in websites Cambridge Dictionary (Cambridge Dictionary[1]), Internet Slangs (Internet Slangs[2]), and Glossary of Video Games Wikipedia page (Glossary of video game terms - Wikipedia[3]) to find out the meanings.

***Note**. Heart emojis (e.g., ♥♥♥♥) in reviews represent censored content. Steam replaces offensive or blocked words (e.g., swear words) with heart emojis but keeps the length of the original word.*

[1] Cambridge Dictionary https://dictionary.cambridge.org/
[2] Internet Slangs https://www.internetslangs.net/
[3] Glossary of Video Games Wikipedia page: https://en.wikipedia.org/wiki/Glossary_of_video_game_terms#B

Description of Units

Context Unit

- The Steam game description, Steam ID, the text of the review, the played hours on record, and the post date.
- Cambridge Dictionary, Internet Slang, and Glossary of Video Games Wikipedia page for clarifying meanings of phrases, slangs, and abbreviations.

Registration Unit

- The entirety of each Steam review

Exclusion Units

- The comments below each review should not be considered.
- Non-English reviews will not be coded, as English is the only common language for all four coders.
- Based on empirical observations during the pilot test, to ensure sufficient context for identifying the specific dimensions mentioned by gamers, a minimum word count of **20** is required. Conversely, to manage the cognitive burden on coders and ensure consistent analysis, reviews exceeding a maximum of **300** words will be excluded.

***Note**. If a review contains an **external link,** coders are instructed to **ignore** any content hosted on the linked page and focus exclusively on the text within the review itself.*

Coding Variables

Administrative Elements

To reiterate, each coder will have an individual coder ID that will be used throughout the coding process. Additionally, coders will identify a unique ID for each review and register the overall recommendation, played hours on record, and the post date as administrative control variables.

V.1 Coder ID

Coder F → 1

Coder H → 2

Coder Q → 3

Coder X → 4

V.2 Production Contexts

Indie → 1

AAA → 2

Instructions:

Please code the identified production contexts which were identified according to the definition below.

Definitions*:*

- Indie games refer to low-budget games produced outside the system of big publishers (Diver, 2016). Indie games are developed by an independent studio that is not owned by or directly affiliated with a major publisher.
    - Indie games are initially identified using Steam tags. For all remaining games, coders verify the employee count through LinkedIn to determine their classification (LinkedIn, n.d.).
- AAA games refer to large-scale commercial games, developed with teams involving many specialized experts, each focusing on different aspects of sound, programming, animation, graphics, marketing, game design, and production. AAA are the games that are published by a major publisher (e.g., Ubisoft, EA, Activision), with a large development team (>100 employees) as sourced from their respective LinkedIn page (Juul, 2019).

- This study utilizes the number of employees of development teams involved in a game's production to classify it as a AAA title. Developer information is sourced from the game's description page, while the specific employee count is verified via the developer's LinkedIn profile.

V.3 Game Genre

Action → 1

Adventure → 2

Strategy → 3

Shooting → 4

Simulation → 5

Instructions:

Please code the identified production contexts as they are identified in the Excel sheet.

Definitions:

- Video game genre labels refer to a set of words or phrases that attempt to describe the experience a game delivers to its potential user base (Simonson et al., 2023).
  - For this study, the researchers first identified eight most prevalent game genres according to literature and empirical observations of popular user-based Steam genre tags: action, adventure, casual, simulation, strategy, shooting, sports, software (Egenfeldt-Nielsen et al., 2015; Lin et al., 2018). The researchers then categorized the 100 games from the "Top 100 Most Played Games" list into these eight genres based on game descriptions and Steam tags. For the final sampling pool, the researchers deleted the genre "software" because such a genre does not constitute games within the conceptual scope of the study. Genres "casual" and "sports" were also excluded, as each contained only one production context (either Indie or AAA), precluding meaningful comparison between the two. From the remaining five game genres, the sample was randomly drawn.

V.4 Review Identification Code

Coders will insert the full identification number code for each review. The review identification code consists of two parts.

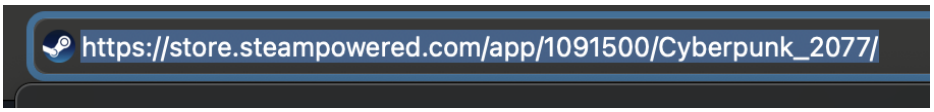
The first part is the Steam code of the game that the reviews refer to. Each game on Steam has a distinct Steam code that makes them identifiable.

For example,

CyberPunk 2077 → 1091500

Figure 1.

Screenshot of Steam URL



As shown in the example above, the Steam code could be found on the link of the game's Steam webpage.

The second part is the assigned index number for each review. The assigned index number for each review is provided to the coder and corresponds to the review's position in the list.

An example of a full identification code would be **1091500_1**.

V.5 Played Hours on Record

"Played hours on record" refers to the amount of time the reviewer had spent playing the game at the time the review was published. Code this variable as a numeral, rounded up to one decimal place. Do not fill in any additional information.

For example:

5.333333 hrs → 5.3

19.8888 hrs → 19.9

V.6 Post Date

The post date refers to the date at which the review is posted. Post date information will be provided to the coders and coded as Year-Month-Date.

For example,

2026-02-17 → 2026-02-17

Main Variables

Coding Instructions for V7, V8, and V9: Coders are to identify the dimensions mentioned within each review.

Please note that **both factual/descriptive mentions and opinionated/evaluative comments qualify as a "Yes"** for these variables. This ensures we record everything when a dimension is brought up, directly supporting our research goal of identifying the most frequently mentioned perspectives.

Coders will first code the sub-elements, followed by the main dimension. Notably, the sub-elements are not comprehensive representations of the main dimension. As some of the review does not necessarily touch a point with any of the sub-elements, but can still be coded "Yes" for the main dimension. Coding the sub-elements first ensures that coders fully grasp the nuances of the dimension and can accurately capture any elements not covered by the sub-elements.

Remarks.

- *Code "No" when the review uses **vague** or **ambiguous** wording that does not clearly specify the perspective being referred to. For example, statements such as "the character is likeable" should be coded as **"No"** as it is unclear whether this refers to visual appearance or personality.*
- *If a term, reference, or phrase remains unclear and relevant definitions cannot be found via the Cambridge Dictionary or internet slang resources or Glossary of Video Games Wikipedia page, for example, capitalized or uncommon words or names, it may refer to in-game context, other related games, films, books, or other publications or media references. Therefore, coders should not rely on the literal meaning of such words when interpreting the review.*
- *Be mindful of any words from the game title appearing in the review, as they may refer to the game itself instead of the literal meaning of the words.*

V7. Aesthetic

V.7.1 Aesthetic - Visual Environment Design

Does the review specifically mention the visual design of the environment or game world?

Yes → 1

No → 0

Instructions: Environment and world design refers to the appearance, layout, and setting of the game's architecture, physical spaces and locations.

Keywords: *World-building, map, world, atmosphere, landscapes, houses, buildings, architecture.*

CODE YES EXAMPLE

"Don't get me wrong, Once Human is a total eye-candy. ***The world looks amazing.*** *But man, the gameplay is a snoozefest. I could easily cheese my way through most stuff, and exploring felt like a chore. And don't even get me started on the player houses blocking everything cool. It's like everyone decided to build their own little fort right in the middle of the action.*

Reason: The review mentions the aesthetics aspects of the world by saying "world looks amazing", as well as characteristic aspects of the world by mentioning that it felt crowded, "houses blocking everything cool. It's like everyone decided to build their own little fort right in the middle of the action". Thus, the coder would code "Yes".

CODE NO EXAMPLE

"There is a very real Subnautica crossed with No Man's Sky vibe thing going on, it all feels very familiar, even the music, this all adds up to something fun, challenging and weirdly familiar all in one."

Reason: From this review, it is unclear what the commenter means by "real," or what aspect they are referring to, whether the game itself, its design, the in-game world, or another feature. The capitalized terms here, *Subnautica* and *No Man's Sky* refer to other games; however, coders should not rely on this literal interpretation. Instead, these references should be ignored, as it cannot be determined whether the reviewer is making a direct comparison or referring to specific elements. Therefore, the comment is too vague and should be coded as "No."

CODE NO EXAMPLE

"Kudos to the visual and audio designers, its a beautiful game and their work is what gives each item and enemy their personality."

Reason: Although the comment praises the visual aspects, it does not specifically refer to the design of the environment. Therefore, this comment should be coded as "No."

V.7.2 Aesthetic - Character Appearance

Does the review specifically mention character appearance (character visual design, modeling)?

Yes → 1

No → 0

Instructions: The visual design and physical rendering of the character/avatar that inhabit the game world. Focus on the facial expressions, visual craftsmanship of the **character**, while ignoring their personality or voice acting.

Keywords: *Character models, outfits, creature design, NPC looks, facial detail, clothes, facial expression, character skins, cute, pretty, attractive, ugly, charming.*

CODE YES EXAMPLE

*"Game got **boobs** and **ass sliders 10/10**"*

Reason*:* Based on the context of the game, there are characters in the game. Although the review is very short, the reference to "boobs" most likely describes the appearance of a character, and the "10/10" at the end indicates that the reviewer is pleased with the character's appearance. Thus, the coder should select "Yes."

CODE YES EXAMPLE

*"its pretty good but its only goonable if you get certain **handler skins** (which you have to pay for)"*

Reason*:* The review mentions "handler skins," and "handler" refers to a type of character in the game. If coders are unsure what "handler" means, they should consult the Cambridge Dictionary for its literal meaning. Therefore, the coder should select "Yes," as it refers to a character's accessory.

CODE NO EXAMPLE

*"The **story is very realistic** and heart touching and the main **characters are likable.** I love **how it was written** and it keeps the reader interested and curious about what will happen later. Once you start it, you can't leave the VN without finishing it. The only thing that effected me negatively is the "violence, harsh themes" that are clearly described in some scenes. but other than this, I give it 10000000 point of 10"*

Reason: This review describes the characters as “likable.” However, “likable” may refer to either personality or appearance, making it ambiguous. Since different coders could interpret this differently, it should be coded as “No.”

V.7.3 Aesthetic - Audio

Does the review mention audio of the game?

Yes → 1

No → 0

Instructions: All audible elements produced by the game, including composed music and functional sound. Focus on how the music and sounds feel and sound.

Keywords: *Score, OST, background music, foley, voiceovers, gun sounds, ambient noise, soundtrack, distracting audio.*

CODE YES EXAMPLE

"This is a must if you ever played Warzone. You see, Warzone mentally scars you while Planet Zoo actively heals your soul. It heals you with its endless positive vibes, ***happy music*** *and cute animals. Baby hippos being my favourite rn."*

Reason: This review mentions that the game "heals you with … happy music", referring that they are pleased with the audio of the game. Thus, the coder selects "yes".

CODE NO EXAMPLE

"Good game, animals good, mechanics good, buildings good. I have spent a total of 45 full days playing this game so far. I do not recommend this but I do recommend the game."

Reason: While this comment mentions game mechanics, which some coders could interpret as all mechanics, including audio, this should be coded as a "no", as it is too ambiguous to place a guess on what exactly is meant.

V.7.4 Aesthetic Dimension (Sobociński, 2019; Wang & Goh, 2020)

Does the review mention any aesthetics aspect of the game?

Yes → 1

No → 0

Instructions: The aesthetics dimension refers to the visual and audio presentation of the game, including how the game looks and sounds. This dimension covers visual environment design, character appearance, overall artistic style, graphic quality and audio. This includes both what the player sees (visuals) and what they hear (audio). Look for any evaluative or descriptive comment regarding the game's sensory output.

Keywords: *looks great, looks bad, sounds immersive, beautiful world, ugly, terrible modelling, skin.*

CODE YES EXAMPLE

"Basically you play as a traumatized space kid who uses their space ninja suits to roam space, meet new people, and kill things alongside other traumatized space kids with spaceninja suits. There's infinite content, ***gorgeous graphics****, an amazing story, wonderful devs &communication and the kindest community I have borne witness to as of yet. We do not deserve this game for free. Cannot recommend it enough."*

Reason: This review mentions many positive aspects of the game, one being "gorgeous graphics" in the game, which is an aesthetics factor. Thus, the coder selects "yes".

CODE NO EXAMPLE

"I like it so far, but there's not a whole lot to do. Like I've already 100%ed all of the levels, but not nearly enough tickets to get the ***cosmetics****. And leveling doesn't do much besides adding more cosmetics. I feel like the game could do with a skill system like Crime Scene Cleaner, like more durable slimes, resist to damage, or maybe some fantastical stuff like double jump boots or a black hole to suck up viscera. Game has potential. 7/10."*

Reason: Even when "cosmetics" are mentioned, they are discussed as a tool in game in terms of unlocking/progression, not their visual design or appeal. Therefore, this review does not mention any aesthetics aspect of the game.

CODE NO EXAMPLE

"I Love Tank Games. Ultra Realistic!!!"

Reason: “Ultra realistic” suggests that the game world is presented in a realistic way. However, it is unclear whether the player is referring to the environment design, character appearance, plot, or the physical realism of gameplay. Therefore, the coder should code “no” for the main dimension.

V8. Narrative

V.8.1 Narrative - Plot

Does the review mention the plot in the game?

Yes → 1

No → 0

Instructions: Plot in narrative refers to **what happens in the story**, in what order the stories as experienced by the player, and the underlying backstory established within the game's world. This includes narrative development, story progression, and **overall story quality**. Notably, quests and missions explicitly framed as related to gameplay do not count. The key is to see how specific parts contribute to the **narrative**.

Keywords: *storytelling, plot, ending, background, story.*

CODE YES EXAMPLE

"Basically you play as a traumatized space kid who uses their space ninja suits to roam space, meet new people, and kill things alongside other traumatized space kids with spaceninja suits. There's infinite content, gorgeous graphics, an ***amazing story****, wonderful devs & communication and the kindest community I have borne witness to as of yet. We do not deserve this game for free. Cannot recommend it enough."*

Reason: The phrase "playing as a traumatized space kid…" appears to describe the game's premise, but in the context of game reviews, such statements usually refer to the **core mechanism** of the game rather than narrative plot events. Plot is only coded when the review explicitly refers to narrative elements such as story, plot, or story progression. In this case, the review is coded "Yes" because it explicitly mentions "an amazing story," which directly refers to the narrative.

CODE NO EXAMPLE

"fun and cool because I raise capybara."

Reason: This review mentions what the player can do in the game "raise capybara" and evaluates the game based on this specific action. However, it does not include any references to the storyline, sequence of events, or narrative progression. Instead, the review focuses more on the game mechanics. This is a borderline case, and coders should code it as "no."

V.8.2 Narrative - Character Build

Does the review mention character build in the game?

Yes → 1

No → 0

Instructions: Character build refers to a character's personalities, personal backstories, growth, **relationships**, or roles within the game world, but is still tied to the character itself, not the overall game plot. Simply mentioning the name of the character is not enough. The review must also mention more in-depth aspects of the character's **personality**, **personal development, motivations, relationships with other characters, or their role in the narrative of the game.** This sub-dimension is different from *Storyline* because it focuses on the **character as an entity** rather than the story events they participate in.

Keywords: *personality, growth, relationship* .

CODE YES EXAMPLE

"...

*Bad: Characters. I wasn't expecting much from an Avatar game in terms of **depth of the characters**, but everyone feels very **tropey**. Occasionally felt like the game didn't know what tone it was going for (campy and one-liner-filled or more serious) Stability. Game crashed a couple of times, making me lose progress. There were also sometimes meshes missing and containers/gatherings I couldn't interact with. The side missions of destroying RDA sites. You've done 3 of them, you've seen it all- Indoor level design. Way too many times I got stuck because it was so unclear what I needed to do/where to go to progress*

..."

Reason: This review gives their opinion on the game while also criticising the character build, as they point out their disappointment in characters by commenting: "I wasn't expecting much from an Avatar game in terms of **depth of the characters**, but everyone feels very tropey". Thus, the coder would code "yes".

CODE NO EXAMPLE

"LET ME ROMANCE SO'LEK"

Reason: While the review is about a character in the game, it is not about how they are like within the game, but about the reviewer's own desires towards the character outside of the game. Thus, this would be coded as a "no" for mentioning character build in the game.

CODE NO EXAMPLE

"I love the look of all mobs, they are very unique and sometimes very funny."

Reason: Although the review mentions "funny," it is clear that this refers to the **appearance rather than the personality** of the mobs. Therefore, the reference is ambiguous and cannot be reliably coded. Hence, this should be coded as "no."

V.8.3 Narrative Dimension (Egenfeldt-Nielsen et al., 2015)

Does the review mention any narrative aspect of the game?

Yes → 1

No → 0

Instructions: "Narrative" can be defined as a succession of events. Its basic components are: the chronological order of the events themselves (story), their written or verbal representation (text), and the act of telling or writing (narration). Narrative factors are anything related to the storytelling elements of the game, including the plot, worldview, characters, storyline, dialogue, and how much the player engages with the story. Look for quotes and descriptions of specific plots or stories in the game as well as overall judgements of the narrative storytelling of the game.

Keywords: *plot, story, storyline, conversation, dialogue, narrative, storytelling, character personality, world setting, world view.*

CODE YES EXAMPLE

"...

*The **story** itself isn't a direct classic Chinese novel "Journey to the West". Instead, it's what-if interpretation of the Monkey King. **This fresh take on the story** makes it feels much more unique and refreshing. While Journey to the West is a **great story**, it has been adapted many times before, so this new take is a welcome change.*

...

*The game is easily 10/10 for me. From it's **immersive storyline** to it's well layed out world-design. This game is definitely your cup of tea if you're into Chinese Mythology or RPG exploration-ish games. It isn't in the same ball-park as Souls-Like game due to it's linear structure and overall more forgiving combat. But the game does have sprinkles of souls-like derived or inspired elements. Highly recommend and I'm excited for their upcoming Black Myth series."*

Reason: This review mentions many narrative aspects of the game, such as "rich use of Chinese mythology", "The story itself…", "from immersive storyline", thus the coder codes "yes".

CODE NO EXAMPLE

"The third person update made this game so much better."

Reason: The word choice of “third person” could seem as a subjective evaluation about the narrative. However, there is no reference to any narrative elements such as story or characters. The wording “third person” most likely refers to a mechanic update. In cases where the references are **vague, unclear or ambiguous**, coders should not guess the meaning and should code “No”, unless narrative aspects are mentioned clearly.

V9. Gameplay

V.9.1 Gameplay - Challenge

Does the review specifically mention the Challenge (Difficulty level) of the game?

Yes → 1

No → 0

***Instructions*:** Challenge refers to how the players review the design of the game's **difficulty level** and the **effort** required by the player to overcome obstacles, or progress through the game. Focus on the difficulty curve, the fairness of obstacles, and the **feeling of overcoming a hurdle**. Rather than the "Hard" controls (that is Control) or "Hard to understand" stories (that is Narrative).

Keywords: *Hard, easy, steep learning curve, accessibility settings, boss difficulty level, fair, frustrating, difficult, challenging, game balance, die a lot, just lose, effort, grinding, grindy, easy to lose.*

CODE YES EXAMPLE

*"whenever i grind for a new weapon and **finally feel im powerful enough the next boss always manages to absolutely vaporize my existence.** so its a peak game would recommend"*

Reason: The phrase "vaporize my existence" indicates repeatedly dying to bosses even after grinding for better equipment. This directly reflects the game's difficulty curve and the challenge of overcoming obstacles, which fits the definition of challenge. Thus, the coder should code "Yes."

CODE YES EXAMPLE

*"Probably one of the best games you can play if you have **patience (you will die a lot)** Strongly recommend playing:)"*

Reason: The review mentions that the player "will die a lot" and that the game requires patience. This reflects the game's difficulty level and the effort needed to progress. Thus, the coder should code "Yes."

CODE YES EXAMPLE

"This game is So good the story is rich and the Cantos Holy they are so FIRE and for like 90% of the game you are going to use Win-rate because why not. But see when its a boss

holy lock the ♥♥♥♥ in *or* ***you just lose*** *best example if you're new you wont get this but for the people who have played the game The middle never forgets"*

Reason: The review mentions that during boss fights, the player must "lock in" or they will "lose". This directly references the difficulty of boss encounters and the level of focus required to succeed. Thus, the coder should code "Yes."

CODE YES EXAMPLE

"The game is way to ***grindy*** *to be fun along with jow money hungry the devs are"*

Reason: The word "grindy" here is a slang that suggests effortful and challenging. Thus, it is coded as "Yes."

CODE NO EXAMPLE

"Played this as someone into Al/ML and it hits different. Great take on ***decision making****,* ***learning****, and breaking out of programmed behaviour without feeling forced. Connor's arc is especially cool from that perspective. Choices actually matter not the fake kind. Kara's story was my favourite. Not about gameplay, more about the experience. One of the best story games I've played."*

Reason: This review focuses on story, character, and player experience related to narrative and decision making. While the reviewer mentions "decision making" and "learning," these refer to the player's personal reflection, not the **effort required to overcome obstacles** or the difficulty of the game's systems. Additionally, phrases like "not about gameplay", "more about the experience" indicate that the reviewer is not discussing the difficulty mechanism. Therefore, the coder should code "no".

CODE NO EXAMPLE

"the best game. thats it. its peak. controls are annoying for beginners tho."

Reason: Annoying does not necessarily mean challenging; it may refer to having many steps or being troublesome to play, rather than being difficult. This could refer to control usability (input difficulty or intuitiveness) rather than gameplay challenge. It concerns how easy the game is to operate, not how hard it is to play in terms of difficulty or skill demand. Thus, the coder should code "no".

CODE NO EXAMPLE

"The **story** is slow-burn but deeply emotional, **rewarding patience** with powerful moments. Gunplay, exploration, and attention to detail set a gold standard for open-world games. A must-play experience that stays with you long after the credits roll."

Reason: Although the review mentions "slow-burn" and "rewarding patience," these refer to the subject of the sentence, which is the "story", narrative pacing, and overall experience, not gameplay challenge or difficulty. Therefore, the coder should code "No." Challenge only applies when the review explicitly or clearly implies difficulty in overcoming gameplay obstacles. "Patience" alone is insufficient.

V.9.2 Gameplay - Control

*Does the review explicitly mention **how the game controls feel or function**?*

Yes → 1

No → 0

Instructions: Control refers to how easily, accurately, and responsively a player can operate the game through inputs from **keyboards, mouse, controllers, and extension softwares** like **mods**. It is **NOT** about **how players control their avatars or storytelling in the game**, nor is it about external technical issues (e.g., Internet connections, laptop performances). It includes input responsiveness (latency/lag), input accuracy (correct or missed actions), control system bugs, keybinding/remapping functionality, keyboard shortcuts, hardware integration issues (keyboard/mouse/controller) within the game, **mods/modding,** and technical bugs occurring in the games.

Keywords: *Responsive, control, keyboards, layout, clunky controls, tight movement, sensitivity, button layout, key remapping, controller support, intuitiveness of movement, technical bugs, tech error, mods.*

CODE YES EXAMPLE

*"...This game was already the best extraction shooter for PVE on the market and it is keeping that position from what I have played. The gunplay is **nice and smooth while giving you a lot of control**..."*

Reason: The review mentions that they enjoy the **smooth** gunplay which gives them a lot of control in the game, referring to how they can operate within the game easily. Thus the coder would code "yes".

CODE YES EXAMPLE

*"...if you wish to give the genre a try, play you some isaac. ive spent countless hours playing and it has **a very large modding scene** so even if you dont want the dlc for the game, its still worth getting as many **mods** shake up the gameplay entirely"*

Reason: This review mentions mods and modding scenes, and should be coded as "Yes."

CODE NO EXAMPLE

"I recently played Detroit: Become Human through once and overall had a really great experience with it. What I especially liked is the story. The decisions you make genuinely feel meaningful and have a clear impact on how the narrative unfolds. This keeps

the game engaging from start to finish, as you are constantly curious about ***how your choices will control the outcome****. The characters are also very well developed, and it was easy to connect with their situations and emotions…"*

Reason: While this review mentions the keyword control, it is talking about player **control over narrative ending** and not about player interaction with the game system. It does not provide any evaluation on button layout, precision. Thus, it is coded as "no."

V.9.3 Gameplay - Core Mechanism

*Does the review specifically mention the **core mechanism?***

Yes → 1

No → 0

Instructions: Core mechanism refers to the **fundamental systems and rules that define how the game is played and what the player does to progress or succeed**. This includes **specific quests, game modes**, combat systems, puzzle-solving mechanics, crafting or resource management systems.

Keywords: *Combat system, game loop, mechanics, puzzle-solving, core mechanism, adventure game, shooting, gameplay mechanism, simulation, rules, PvP, multi-player, game modes, playing with a friend, open-world, gunplay.*

CODE YES EXAMPLE

*"It's not for action lovers, as gameplay can feel slow and repetitive over time, but if you enjoy **simulation** and **peaceful exploration**, it's highly satisfying"*

Reason: The review reflects on the core gameplay experience by referencing game modes such as **simulation and exploration**. Therefore, the coder should select "Yes."

CODE YES EXAMPLE

*"fun ivolved game. the **people you play with can often impact the experience**, you need to be the difference."*

Reason: The review reflects on the social and **interactive** core mechanism of gameplay by highlighting how **other players influence the game**. Therefore, the coder should select "Yes."

CODE NO EXAMPLE

"I miss when games launched with all content included."

Reason: This review is about the players general frustrations towards the gaming industry and business practices of the developers, however **no references to the specific gameplay or mechanic systems** are made, meaning on how this affects the way that the game is played (making the experience more challenging for example). Thus, this would be coded as a "No".

V.9.4 Gameplay Dimension (Sobociński, 2019; Zagal et al., 2012)

Does this review mention Gameplay?

Yes → 1

No → 0

***Instructions*:** Gameplay refers to the player interaction with the game system and with the other players in the game. Focus on **interaction with the games' system**, **technical bugs from the game's side**, **specific game modes**, core mechanics, rules and regulations of the games and the interaction with other players inside the games. Mentioning of interactions with game community members on streaming channels outside the game does not count.

Keywords: *Gameplay, mechanics, systems, balance, difficulty, challenge, controls, interface, modes, PvP, Co-op, grind, repetitive, loop, addictive, interaction, playability, bugs.*

CODE YES EXAMPLE

*"...The **balancing in this game** is just awful. The devs clearly know there is a problem of no one wanting to play Vanguard, but instead of incentivising the players to play tank, they make tanking an awful experience.*

..."

Reason: This review mentions a keyword regarding gameplay which is **balancing**, and marking that problems in this make playing the game an "awful experience". This refers to how game mechanics are tuned and experienced in play, which is a core aspect of gameplay. Therefore, the coder should select "Yes."

CODE NO EXAMPLE

"..recommend watch some gameplay before playing."

Reason: The coders need to take into account the different wordings, context and phrases used by gamers in their reviews. While this review mentions the keyword "gameplay," it only recommends viewing gameplay footage, which is irrelevant to the gameplay dimension. Thus, it is coded "no."

Coder Remark

Do you have any coder remarks regarding this review?

Please provide your remarks here: _____

Instructions: Leave a remark only if you are extremely concerned and have serious observations, uncertainties, or difficulties encountered while coding this review. This may include unclear content, challenges in applying coding categories, or aspects of the review that were not fully captured by the existing coding instructions. If relevant, you may also suggest potential improvements to the codebook.

[End of codebook]

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